XRPL Validator — Daily Engineering Report · 2026-07-04

Period: 2026-07-04 00:00 to 24:00 US Eastern (2026-07-04 04:00 to 2026-07-05 04:00 UTC). Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: xrpl-validator merge records (they begin 2026-07-04); data.xrpl.org agreement scores. Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); amendment data.

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** no receipt records for this day (they begin 2026-08-29).
- **Changes:** 0 deployments, 47 commits merged into main.

Service levels

No signing records for this day (they begin 2026-09-19).

External agreement, data.xrpl.org (UTC day 2026-07-04): 0.99511 (105 missed of 21,489). *data.xrpl.org scores UTC days from the validations its crawlers received.*

Native engine conformance

No receipt records for this day (Step 1 records begin 2026-08-29, Steps 2–5 records 2026-10-09).

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipt records for this day (Step 1 records begin 2026-08-29).

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

Merged	Repo	Commit	Subject
16:04 UTC	xrpl-validator	`afa3896`	feat(dashboard): copilot investigate panel + ignore copilot/ backend
16:04 UTC	xrpl-validator	`1c66cf1`	fix(dashboard): bump LIBXRPL badge 3.1.3 -> 3.2.0
16:04 UTC	xrpl-validator	`7e5b1b9`	fix(peer): raise codec MAX_MESSAGE_SIZE 4MiB -> 64MiB to match rippled 3.2.0
16:04 UTC	xrpl-validator	`294e85c`	feat(lockstep): expose xrpl_lockstep_shadow_match_ratio gauge (M1 task 5)
16:04 UTC	xrpl-validator	`3ff0144`	feat(lockstep): env-gated ledger_hash shadow eval at the FFI hook (M1 task 4)
16:04 UTC	xrpl-validator	`13e6ecc`	fix(ffi): default HashRouter::Setup (setup_HashRouter moved to xrpld in 3.2.0)
16:04 UTC	xrpl-validator	`b9cf3ce`	fix(ffi): NoopOrderBookDB for 3.2.0 (OrderBookDB now pure-virtual)
16:04 UTC	xrpl-validator	`2f0f58c`	fix(ffi): 3.2.0 header paths for HashRouter/LoadFeeTrack/OrderBookDB
16:04 UTC	xrpl-validator	`db47945`	fix(ffi): make getHashRouter/getFeeTrack/getOrderBookDB/getLogs real for 3.2.0
16:04 UTC	xrpl-validator	`15f602f`	fix(ffi): MinimalApp::getNetworkIDService real (NetworkIDServiceImpl) for 3.2.0
16:04 UTC	xrpl-validator	`92ca532`	feat(validator): API bind port is FFI_PORT-configurable (default 3777)
16:04 UTC	xrpl-validator	`1ecb3f5`	fix(ffi): beast::Severity enum values are Fatal/Error (not kFatal/kError)
16:04 UTC	xrpl-validator	`6b77e3c`	fix(ffi): MinimalApp implements all 46 ServiceRegistry virtuals + xrpl_shim 3.2.0
16:04 UTC	xrpl-validator	`7e13344`	fix(ffi): rewrite MinimalApp for 3.2.0's 16-virtual Application + shim fixes
16:04 UTC	xrpl-validator	`99c45d6`	fix(ffi): repath MinimalApp includes + digest type for 3.2.0 libxrpl/xrpld reorg
16:04 UTC	xrpl-validator	`c166a21`	fix(ffi): adapt CallbackReadView to rippled 3.2.0 ReadView interface

Merged	Repo	Commit	Subject
16:04 UTC	xrpl-validator	`6ef2393`	fix(ffi): migrate shim from namespace ripple to xrpl for rippled 3.2.0
16:04 UTC	xrpl-validator	`7238d72`	fix(ffi): port shim build to rippled 3.2.0 — ed25519/secp256k1 now conan pkgs
16:04 UTC	xrpl-validator	`7f15b90`	feat(validator): regenerate amendment vote list for rippled 3.2.0
16:04 UTC	xrpl-validator	`8bb6a6f`	docs(amendments): read-only vote-board design spec (v1)
16:04 UTC	xrpl-validator	`c50394e`	chore(3.2.0-prep): F8c shim version comments + port plan + release watch
16:04 UTC	xrpl-validator	`2279891`	style(validator): use u32::is_multiple_of in is_flag_ledger (clippy)
16:04 UTC	xrpl-validator	`826406e`	fix(ledger): preserve leaf data on SHAMap delete collapse (reaudit F4 / val-110)
16:04 UTC	xrpl-validator	`ecf3ebc`	feat(validator): refresh active amendments on flag ledgers (reaudit F6/va-07)
16:04 UTC	xrpl-validator	`28f6658`	feat(validator): encode crash-consistency contract via clean-shutdown marker (reaudit F5)
16:04 UTC	xrpl-validator	`7501eab`	fix(validator): refuse signing startup with untrusted RPC endpoints (reaudit F7)
16:04 UTC	xrpl-validator	`a99dbb1`	fix(validator): remove dead full-DB sweep footgun (reaudit F2)
16:04 UTC	xrpl-validator	`de6ea4b`	chore(validator): commit ffi_test sidecar arg drift to extended apply_ledger_in_order signature (reaudit F8a)
16:04 UTC	xrpl-validator	`2d471f3`	fix(validator): disarm va-05 hard signing gate by default (XRPL_VA05_ENFORCE)
16:04 UTC	xrpl-validator	`ccc0fc6`	feat(lockstep): carry close_time/resolution/flags/ledger_hash in LedgerHeader (M1 task 3)
16:04 UTC	xrpl-validator	`e19a27f`	feat(lockstep): rolling match-ratio meter (M1 task 2)
16:04 UTC	xrpl-validator	`201ed9e`	feat(lockstep): pure ledger-hash evaluation (M1 task 1)
16:04 UTC	xrpl-validator	`b5b963e`	docs(lockstep): M1 shadow-harness implementation plan
16:04 UTC	xrpl-validator	`40d596c`	docs(lockstep): M1 shadow-harness design spec
16:04 UTC	xrpl-validator	`f003368`	feat(validator): sign locally-computed ledger_hash; refuse on network mismatch
16:04 UTC	xrpl-validator	`6213142`	Merge main (MIT relicense + IP scrub) into feat/state-rocks-miss-buckets
16:04 UTC	xrpl-validator	`6687d20`	feat(the validator's ledger feed): tx-set completeness gate to prevent transient-pull halts
16:04 UTC	xrpl-validator	`70f6c0d`	fix(the validator's ledger feed): halt-on-mismatch + write-after-verify + lag breadcrumbs
16:04 UTC	xrpl-validator	`7f66b81`	feat(xrpl-node): add state.rocks read-latency histograms + raise consult budget to 500
16:04 UTC	xrpl-validator	`7efb896`	feat(xrpl-node): bucket state.rocks misses by RPC outcome
16:04 UTC	xrpl-validator	`551e804`	chore(node): ffi_test — match apply_ledger_in_order 4 new optional log/expectation params

Merged	Repo	Commit	Subject
16:04 UTC	xrpl-validator	`3986917`	merge: feat/state-rocks-miss-buckets (the validator host tip) into main (state-rocks miss instrumentation, ffi_test signature chore)
16:04 UTC	xrpl-validator	`f995252`	merge: fix/succ-exclusive-bound into main (6th-halt NFTokenPage fix: succ open interval)
16:04 UTC	xrpl-validator	`59bcaec`	fix(ffi): make CallbackReadView succ upper bound exclusive (open interval)
16:04 UTC	xrpl-validator	`6893bbe`	fix(ffi): make CallbackReadView succ upper bound exclusive (open interval)
16:09 UTC	xrpl-validator	`328f089`	chore(ffi): SHIM_VERSION 0.3.0 — 3.2.0 shim port; docs: STATE-2026-07 current-truth (M1 sign-off + stale-claim corrections)
16:22 UTC	xrpl-validator	`03af266`	feat(node): S1/S2 peer signature verification — proposals + validations (observe-mode)

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **feat(dashboard): copilot investigate panel + ignore copilot/ backend** (`afa3896` , xrpl-validator) Embeds the copilot investigate panel in the dashboard (live on :3778); next.config proxies to the copilot service. The /copilot backend is its own repo, so gitignore it here.
- **fix(dashboard): bump LIBXRPL badge 3.1.3 -> 3.2.0** (`1c66cf1` , xrpl-validator) Validator runs libxrpl 3.2.0; header badge was a stale hardcoded literal.
- **fix(peer): raise codec MAX_MESSAGE_SIZE 4MiB -> 64MiB to match rippled 3.2.0** (`7e5b1b9` , xrpl-validator) 3.2.0 peers legitimately send bulk frames up to kMaximumMessageSize=megabytes(64) (overlay/Message.h; 26-bit payload header field). The stale 4 MiB cap silently dropped ~60 MB messages from many peers (116 codec errors observed post-upgrade). Not consensus-critical (state syncs from the reference node, not peer bulk) but correct protocol behavior. Tests: oversized boundary now 65 MiB; new test asserts sub-cap accept.
- **feat(lockstep): expose xrpl_lockstep_shadow_match_ratio gauge (M1 task 5)** (`294e85c` , xrpl-validator)
- **feat(lockstep): env-gated ledger_hash shadow eval at the FFI hook (M1 task 4)** (`3ff0144` , xrpl-validator) check_shadow_hash now returns the computed state root (Option<[u8;32]>) instead of bool; the ws_sync FFI-shadow hook feeds it + header fields into lockstep::evaluate_ledger when XRPL_LOCKSTEP_SHADOW=1, recording a process-wide LockstepMeter. Shadow-only, no signing/state impact.
- **fix(ffi): default HashRouter::Setup (setup_HashRouter moved to xrpld in 3.2.0)** (`13e6ecc` , xrpl-validator)
- **fix(ffi): NoopOrderBookDB for 3.2.0 (OrderBookDB now pure-virtual)** (`b9cf3ce` , xrpl-validator) 3.2.0 turned OrderBookDB into a 5-method interface; the concrete OrderBookDBImpl needs a full ServiceRegistry. Single-tx apply crosses offers via the ledger View's book-directory SLEs, not this cache, so a no-op impl is hash-safe.
- **fix(ffi): 3.2.0 header paths for HashRouter/LoadFeeTrack/OrderBookDB** (`2f0f58c` , xrpl-validator) 3.2.0 relocated these out of xrpld/app/*: HashRouter->xrpl/core, LoadFeeTrack->xrpl/server, OrderBookDB->xrpl/ledger.

- **fix(ffi): make getHashRouter/getFeeTrack/getOrderBookDB/getLogs real for 3.2.0** (`db47945` , xrpl-validator) Port the <=3.1.x shim's real service impls to the 3.2.0 MinimalApp: HashRouter (setup_HashRouter + stopwatch), LoadFeeTrack (null journal), OrderBookDB(*this), and Logs(Fatal) that OrderBookDB needs. The apply()/preflight() chain reaches these for regular tx types; previously throw-stubbed -> every live-apply hit tefEXCEPTION (getHashRouter). Signing path is unaffected (independent of FFI).
- **fix(ffi): MinimalApp::getNetworkIDService real (NetworkIDServiceImpl) for 3.2.0** (`15f602f` , xrpl-validator) The shadow showed every FFI apply throwing tefEXCEPTION 'MinimalApp::getNetworkIDService — not implemented'. 3.2.0 preflight routes the tx NetworkID check through getNetworkIDService(); implement it like the real Application: make_unique(config.networkId).
- **feat(validator): API bind port is FFI_PORT-configurable (default 3777)** (`92ca532` , xrpl-validator) Lets a shadow/test instance run beside the live validator on a different port (FFI_PORT=3787); production unchanged (default 3777).
- **fix(ffi): beast::Severity enum values are Fatal/Error (not kFatal/kError)** (`1ecb3f5` , xrpl-validator)
- **fix(ffi): MinimalApp implements all 46 ServiceRegistry virtuals + xrpl_shim 3.2.0** (`6b77e3c` , xrpl-validator) 3.2.0 moved the service accessors into a ServiceRegistry base (46 pure virtuals). MinimalApp now implements all 62 (16 Application + 46 ServiceRegistry): config() real (networkId); isStopping/getApp/getJournal(null sink)/getTrapTxID kept safe so apply-path lifecycle/logging calls don't throw; the rest are throw-stubs (shadow window will reveal any apply actually needs — e.g. …
- **fix(ffi): rewrite MinimalApp for 3.2.0's 16-virtual Application + shim fixes** (`7e13344` , xrpl-validator) 3.2.0 slimmed Application to 16 pure virtuals (lifecycle + config/nodeIdentity/etc; heavy service accessors gone). Rewrote MinimalApp to implement exactly those 16 (config() real w/ networkId; rest stub-throw), dropping ~35 obsolete overrides + the hashRouter/loadFeeTrack/orderBookDB/logs members — which also clears the io_service, PathRequests, OrderBookDB-incomplete, setup_HashRouter, beast::severities, and …
- **fix(ffi): repath MinimalApp includes + digest type for 3.2.0 libxrpl/xrpld reorg** (`99c45d6` , xrpl-validator) 3.2.0 moved headers from xrpld/ into libxrpl: HashRouter.h xrpld/app/misc->xrpl/core; LoadFeeTrack.h xrpld/app/misc->xrpl/server; Ledger.h xrpld/app/ledger->xrpl/ledger. CallbackReadView::digest now returns std::optional (digest_type alias not member-accessible). Application/Config/OrderBookDB stayed in xrpld.
- **fix(ffi): adapt CallbackReadView to rippled 3.2.0 ReadView interface** (`c166a21` , xrpl-validator) info()->header() (LedgerInfo->LedgerHeader); sles_type/txs_type renamed SlesType/ TxsType; balanceHook split into balanceHookIOU/MPT/SelfIssueMPT (non-pure) + ownerCountHook non-pure -> drop no-op overrides; digest() now pure -> throw-stub.
- **fix(ffi): migrate shim from namespace ripple to xrpl for rippled 3.2.0** (`6ef2393` , xrpl-validator) 3.2.0 completed the ripple->xrpl namespace rename across libxrpl/xrpld (no compat alias). Migrate the hand-written shim files (CallbackReadView/MinimalApp/ MutationCollector + xrpl_shim.cpp): namespace decls + all ripple:: qualifiers -> xrpl.
- **fix(ffi): port shim build to rippled 3.2.0 — ed25519/secp256k1 now conan pkgs** (`7238d72` , xrpl-validator) 3.2.0 moved ed25519/secp256k1 from external/ source builds to conan packages (targets ed25519::ed25519 / secp256k1::secp256k1, per rippled cmake/XrplCore.cmake). CMakeLists: drop dead external/-path IMPORTED targets -> find_package + conan targets. build.rs: drop manual ed25519/secp256k1 link lines (collect_conan_libs resolves them).
- **feat(validator): regenerate amendment vote list for rippled 3.2.0** (`7f15b90` , xrpl-validator) Regenerated supported_amendments() from 3.2.0 features.macro (Supported::Yes, 46): drops the amendments 3.2.0 retired into

baseline, adds the new supported 3.2.0 amendments (LendingProtocol/TokenEscrow/PermissionedDEX/SingleAssetVault/DynamicNFT/ MPTokensV1 + fixes), and excludes the 5 Supported::No placeholders (MPTokensV2/Batch/ PermissionDelegationV1_1/DynamicMPT/fixBatchInnerSigs) the binary can't yet apply. …

- **fix(ledger): preserve leaf data on SHAMap delete collapse (reaudit F4 / val-110)** (`826406e`, xrpl-validator) delete_from_mut's structural-collapse path promoted the surviving leaf via find_sole_leaf -> LeafNode::new_hash_only(key, hash), which zeroes data (data: Vec::new()). In a State tree the surviving sibling kept its hash but lost its bytes — invisible to the hash-based signing path, but lookup() returned Some([]). This blocked lockstep M1 and any path trusting SHAMap::delete.
- **feat(validator): refresh active amendments on flag ledgers (reaudit F6/va-07)** (`ecf3ebc`, xrpl-validator) The FFI verifier fetched the active mainnet amendment set once at startup and froze it for the process lifetime. An amendment that reaches 2-week majority and activates while the validator runs would be absent → every tx exercising it diverges → 3 strikes → halt; this fires more often in the post-3.2.0 wave.
- **feat(validator): encode crash-consistency contract via clean-shutdown marker (reaudit F5)** (`28f6658`, xrpl-validator) The 'wipe state.rocks + bulk-resync on unclean restart' contract lived only in operator discipline. A clean shutdown now writes clean_shutdown.marker (SIGTERM handler, after the leaf-cache save); boot consumes it. Marker present => prior shutdown clean, state consistent, resume. Marker absent while state.rocks exists => unclean (OOM/SIGKILL/power) => state may be torn (hasher rebuilds from inconsistent rocks → …
- **fix(validator): refuse signing startup with untrusted RPC endpoints (reaudit F7)** (`7501eab`, xrpl-validator) A signing validator on a host with no local rippled (the validator host) silently fails over from localhost to the public clusters (xrplcluster.com / s1.ripple.com) when XRPL_RPC_URL/XRPL_WS_URL are unset — a consensus-divergence source and a violation of the never-hammer-public-RPCs rule. The binary never enforced the operational 'MUST set both env vars' rule.
- **fix(validator): remove dead full-DB sweep footgun (reaudit F2)** (`a99dbb1`, xrpl-validator) start_sweep + fetch_batch re-fetched every state.rocks key at one pinned ledger and wrote each with an unconditional db.put NOT paired with a hasher update. Run alongside live sync it regressed freshly-synced keys to stale pinned-seq values while the in-memory hasher kept the live value — silent rocks rot that detonates on the next restart (hasher rebuilt from rocks → wrong account_hash → 3 strikes → halt). Zero …
- **fix(validator): disarm va-05 hard signing gate by default (XRPL_VA05_ENFORCE)** (`2d471f3`, xrpl-validator) Reaudit 2026-06-10 finding F1 (SEV-1): the va-05 block was live-wired unconditionally, so any rebuild from this branch (3.2.0 forces one) would re-arm the gate that failed on 2026-05-31 — the validator's ledger feed's account_hash trails the live edge, pairing account_hash(N-lag) with header(N), mismatching every ledger and refusing all signing.
- **feat(lockstep): carry close_time/resolution/flags/ledger_hash in LedgerHeader (M1 task 3)** (`ccc0fc6`, xrpl-validator)
- **feat(lockstep): rolling match-ratio meter (M1 task 2)** (`e19a27f`, xrpl-validator)
- **feat(lockstep): pure ledger-hash evaluation (M1 task 1)** (`201ed9e`, xrpl-validator)
- **feat(validator): sign locally-computed ledger_hash; refuse on network mismatch** (`f003368`, xrpl-validator) VALAUDIT Phase 5 (va-05). After the phase-3 gate opens, recompute the full ledger_hash from the closed-ledger header fields + our independently-computed account_hash, and sign it only if it equals the network's — the signature now proves independent verification of that ledger. Refuse (fail-closed) on any divergence or a missing/unfetchable header. Header sourced via an RPC `ledger` fetch from local rippled at sign …

- **feat(the validator's ledger feed): tx-set completeness gate to prevent transient-pull halts** (`6687d20` , xrpl-validator) Recompute each pulled ledger's transaction-tree root and compare to the header transaction_hash BEFORE applying. An incomplete RPC pull (root cause of the 2026-05-28 halt) is then re-driven instead of producing a spurious state-hash divergence that counts toward halt-on-mismatch.
- **fix(the validator's ledger feed): halt-on-mismatch + write-after-verify + lag breadcrumbs** (`70f6c0d` , xrpl-validator) Three layered fixes for the failure mode that crashed the validator on 2026-05-19: the validator's ledger feed silently lagged 51 ledgers behind the live edge, the watchdog forced a reconnect, recovery couldn't catch up, and the validator produced 263 consecutive state-hash MISMATCHes before dying. The pre-fix "log and move on" behaviour committed corrupt state.rocks data once and re-corrupted it every ledger after.
- **feat(xrpl-node): add state.rocks read-latency histograms + raise consult budget to 500** (`7f66b81` , xrpl-validator) Two changes that together make the bucketing instrumentation a permanent canary for state.rocks rot (the failure mode from 2026-05-15).
- **feat(xrpl-node): bucket state.rocks misses by RPC outcome** (`7efb896` , xrpl-validator) When state.rocks misses, consult rippled (under a per-ledger budget) and bucket the miss by what rippled said: - legitimate_miss: rippled also said entryNotFound (bloom filter candidate) - should_exist_miss: rippled returned bytes, state.rocks has a gap (read-through cache candidate) - rpc_error: retries exhausted, outcome unknown
- **fix(ffi): make CallbackReadView succ upper bound exclusive (open interval)** (`59bcaec` , xrpl-validator) rippled's ReadView::succ contract is the OPEN interval (key, last). The OverlayedDbProvider::succ callback treated `last` as inclusive (k <= last) in both the snapshot scan and the overlay filter. Dormant for owner-dir / order-book walks (their `last` sentinel never equals a live key) but load-bearing for NFTokenMint: rippled locates a mint's target page via succ(first, nftpageMax(owner).next()), so an inclusive …
- **fix(ffi): make CallbackReadView succ upper bound exclusive (open interval)** (`6893bbe` , xrpl-validator) rippled's ReadView::succ contract is the OPEN interval (key, last). The OverlayedDbProvider::succ callback treated `last` as inclusive (k <= last) in both the snapshot scan and the overlay filter. Dormant for owner-dir / order-book walks (their `last` sentinel never equals a live key) but load-bearing for NFTokenMint: rippled locates a mint's target page via succ(first, nftpageMax(owner).next()), so an inclusive …
- **feat(node): S1/S2 peer signature verification — proposals + validations (observe-mode)** (`03af266` , xrpl-validator) Closes SECURITY 7.3/7.4 measurement gap: TMProposeSet signatures verified over sha512Half('PRP\0' || be32(proposeSeq) || be32(closeTime) || prevLedger || txSetHash) (secp256k1 digest-verify, ed25519 counted unsupported per rippled verifyDigest); STValidation blobs field-walked, Signature stripped, 'VAL\0' preimage verified with key-type dispatch. Counters on /metrics (xrpl_proposal|validation_sig_*_total); …

Docs and operations

- **docs(amendments): read-only vote-board design spec (v1)** (`8bb6a6f` , xrpl-validator)
- **chore(3.2.0-prep): F8c shim version comments + port plan + release watch** (`c50394e` , xrpl-validator)
- **style(validator): use u32::is_multiple_of in is_flag_ledger (clippy)** (`2279891` , xrpl-validator)
- **chore(validator): commit ffi_test sidecar arg drift to extended apply_ledger_in_order signature (reaudit F8a)** (`de6ea4b` , xrpl-validator)
- **docs(lockstep): M1 shadow-harness implementation plan** (`b5b963e` , xrpl-validator)
- **docs(lockstep): M1 shadow-harness design spec** (`40d596c` , xrpl-validator)
- **Merge main (MIT relicense + IP scrub) into feat/state-rocks-miss-buckets** (`6213142` , xrpl-validator)

- **chore(node): ffi_test — match apply_ledger_in_order 4 new optional log/expectation params** (`551e804`, xrpl-validator)
- **merge: feat/state-rocks-miss-buckets (the validator host tip) into main (state-rocks miss instrumentation, ffi_test signature chore)** (`3986917`, xrpl-validator)
- **merge: fix/succ-exclusive-bound into main (6th-halt NFTokenPage fix: succ open interval)** (`f995252`, xrpl-validator)
- **chore(ffi): SHIM_VERSION 0.3.0 — 3.2.0 shim port; docs: STATE-2026-07 current-truth (M1 sign-off + stale-claim corrections)** (`328f089`, xrpl-validator)

Independence program

No independence record for UTC day 2026-07-04; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

No amendment records for this day.

Open items

None.

Corrections to earlier reports

None.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Oracle Auto-Settlement for Cash Options** (amm-defi) Extend XLS-62's OptionCreate to accept a tfCashSettled flag plus sfSettlementOracleID (a PriceOracle keylet reference); the writer's collateral is loc
- **Native Structured Metadata for Tokens** (tokens) Add a new owner-reserved ledger object ltTOKENMETADATA at keylet::tokenMetadata(issuer, currency-or-MPTIssuanceID), created and updated by a new issue

- **Revocable On-Ledger Attestation Registry** (audit) Add a new owner-reserved ledger object ltATTESTATION at keylet::attestation(attester, sfAttestationID), where sfAttestationID is an attester-chosen la

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
06-20 15:34	07-04 16:04	xrpl-validator	`afa3896`	feat(dashboard): copilot investigate panel + ignore copilot/ backend
06-20 15:34	07-04 16:04	xrpl-validator	`1c66cf1`	fix(dashboard): bump LIBXRPL badge 3.1.3 -> 3.2.0
06-16 16:27	07-04 16:04	xrpl-validator	`7e5b1b9`	fix(peer): raise codec MAX_MESSAGE_SIZE 4MiB -> 64MiB to match rippled 3.2.0
06-16 16:13	07-04 16:04	xrpl-validator	`294e85c`	feat(lockstep): expose xrpl_lockstep_shadow_match_ratio gauge (M1 task 5)
06-16 16:13	07-04 16:04	xrpl-validator	`3ff0144`	feat(lockstep): env-gated ledger_hash shadow eval at the FFI hook (M1 task 4)
06-16 04:06	07-04 16:04	xrpl-validator	`13e6ecc`	fix(ffi): default HashRouter::Setup (setup_HashRouter moved to xrpld in 3.2.0)
06-16 04:05	07-04 16:04	xrpl-validator	`b9cf3ce`	fix(ffi): NoopOrderBookDB for 3.2.0 (OrderBookDB now pure-virtual)
06-16 04:03	07-04 16:04	xrpl-validator	`2f0f58c`	fix(ffi): 3.2.0 header paths for HashRouter/LoadFeeTrack/OrderBookDB
06-16 03:59	07-04 16:04	xrpl-validator	`db47945`	fix(ffi): make getHashRouter/getFeeTrack/getOrderBookDB/getLogs real for 3.2.0
06-16 03:11	07-04 16:04	xrpl-validator	`15f602f`	fix(ffi): MinimalApp::getNetworkIDService real (NetworkIDServiceImpl) for 3.2.0
06-16 03:02	07-04 16:04	xrpl-validator	`92ca532`	feat(validator): API bind port is FFI_PORT-configurable (default 3777)
06-16 02:36	07-04 16:04	xrpl-validator	`1ecb3f5`	fix(ffi): beast::Severity enum values are Fatal/Error (not kFatal/kError)
06-16 02:34	07-04 16:04	xrpl-validator	`6b77e3c`	fix(ffi): MinimalApp implements all 46 ServiceRegistry virtuals + xrpl_shim 3.2.0
06-16 02:26	07-04 16:04	xrpl-validator	`7e13344`	fix(ffi): rewrite MinimalApp for 3.2.0's 16-virtual Application + shim fixes
06-16 02:15	07-04 16:04	xrpl-validator	`99c45d6`	fix(ffi): repath MinimalApp includes + digest type for 3.2.0 libxrpl/xrpld reorg
06-16 02:10	07-04 16:04	xrpl-validator	`c166a21`	fix(ffi): adapt CallbackReadView to rippled 3.2.0 ReadView interface
06-16 02:02	07-04 16:04	xrpl-validator	`6ef2393`	fix(ffi): migrate shim from namespace ripple to xrpl for rippled 3.2.0
06-16 01:56	07-04 16:04	xrpl-validator	`7238d72`	fix(ffi): port shim build to rippled 3.2.0 — ed25519/secp256k1 now conan pkgs
06-16 01:28	07-04 16:04	xrpl-validator	`7f15b90`	feat(validator): regenerate amendment vote list for rippled 3.2.0
06-15 13:41	07-04 16:04	xrpl-validator	`8bb6a6f`	docs(amendments): read-only vote-board design spec (v1)
06-15 12:09	07-04 16:04	xrpl-validator	`c50394e`	chore(3.2.0-prep): F8c shim version comments + port plan + release watch
06-14 14:10	07-04 16:04	xrpl-validator	`2279891`	style(validator): use u32::is_multiple_of in is_flag_ledger (clippy)

Committed	Merged into main	Repo	Commit	Subject
06-14 13:53	07-04 16:04	xrpl-validator	`826406e`	fix(ledger): preserve leaf data on SHAMap delete collapse (reaudit F4 / val-110)
06-14 13:47	07-04 16:04	xrpl-validator	`ecf3ebc`	feat(validator): refresh active amendments on flag ledgers (reaudit F6/va-07)
06-14 13:34	07-04 16:04	xrpl-validator	`28f6658`	feat(validator): encode crash-consistency contract via clean-shutdown marker (reaudit F5)
06-14 13:28	07-04 16:04	xrpl-validator	`7501eab`	fix(validator): refuse signing startup with untrusted RPC endpoints (reaudit F7)
06-14 13:21	07-04 16:04	xrpl-validator	`a99dbb1`	fix(validator): remove dead full-DB sweep footgun (reaudit F2)
06-14 13:10	07-04 16:04	xrpl-validator	`de6ea4b`	chore(validator): commit ffi_test sidecar arg drift to extended apply_ledger_in_order signature (reaudit F8a)
06-11 02:47	07-04 16:04	xrpl-validator	`2d471f3`	fix(validator): disarm va-05 hard signing gate by default (XRPL_VA05_ENFORCE)
05-31 18:14	07-04 16:04	xrpl-validator	`ccc0fc6`	feat(lockstep): carry close_time/resolution/flags/ledger_hash in LedgerHeader (M1 task 3)
05-31 18:12	07-04 16:04	xrpl-validator	`e19a27f`	feat(lockstep): rolling match-ratio meter (M1 task 2)
05-31 18:09	07-04 16:04	xrpl-validator	`201ed9e`	feat(lockstep): pure ledger-hash evaluation (M1 task 1)
05-31 17:50	07-04 16:04	xrpl-validator	`b5b963e`	docs(lockstep): M1 shadow-harness implementation plan
05-31 17:45	07-04 16:04	xrpl-validator	`40d596c`	docs(lockstep): M1 shadow-harness design spec
05-31 16:54	07-04 16:04	xrpl-validator	`f003368`	feat(validator): sign locally-computed ledger_hash; refuse on network mismatch
05-31 14:51	07-04 16:04	xrpl-validator	`6213142`	Merge main (MIT relicense + IP scrub) into feat/state-rocks-miss-buckets
05-29 05:50	07-04 16:04	xrpl-validator	`6687d20`	feat(the validator's ledger feed): tx-set completeness gate to prevent transient-pull halts
05-19 18:38	07-04 16:04	xrpl-validator	`70f6c0d`	fix(the validator's ledger feed): halt-on-mismatch + write-after-verify + lag breadcrumbs
05-18 00:33	07-04 16:04	xrpl-validator	`7f66b81`	feat(xrpl-node): add state.rocks read-latency histograms + raise consult budget to 500
05-17 01:03	07-04 16:04	xrpl-validator	`7efb896`	feat(xrpl-node): bucket state.rocks misses by RPC outcome
07-04 16:02	07-04 16:04	xrpl-validator	`551e804`	chore(node): ffi_test — match apply_ledger_in_order 4 new optional log/expectation params
07-04 16:04	07-04 16:04	xrpl-validator	`3986917`	merge: feat/state-rocks-miss-buckets (the validator host tip) into main (state-rocks miss instrumentation, ffi_test signature chore)
07-04 16:04	07-04 16:04	xrpl-validator	`f995252`	merge: fix/succ-exclusive-bound into main (6th-halt NFTokenPage fix: succ open interval)
06-30 22:08	07-04 16:04	xrpl-validator	`59bcaec`	fix(ffi): make CallbackReadView succ upper bound exclusive (open interval)
06-30 21:51	07-04 16:04	xrpl-validator	`6893bbe`	fix(ffi): make CallbackReadView succ upper bound exclusive (open interval)
07-04 16:09	07-04 16:09	xrpl-validator	`328f089`	chore(ffi): SHIM_VERSION 0.3.0 — 3.2.0 shim port; docs: STATE-2026-07 current-truth (M1 sign-off + stale-claim corrections)
07-04 16:22	07-04 16:22	xrpl-validator	`03af266`	feat(node): S1/S2 peer signature verification — proposals + validations (observe-mode)

Times in UTC. A commit appears on the day it was committed to main or merged into it.